

CMPT 305 Project Report - Group 9

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Results

Trace	Depth	Replication	Cycles	Execution Time (ms)	Integer (%)	FP (%)	Branch (%)	Load (%)	Store (%)
srv_0	1	1	1408889	1.408889	47.5764	0.4487	18.7246	22.9315	10.3188
srv_0	1	2	1379185	1.379185	47.9605	0.7024	18.2051	22.8653	10.2667
srv_0	1	3	1396131	1.396131	47.7866	0.5066	18.4880	22.8909	10.3279
srv_0	1	4	1388349	1.388349	47.6417	0.6309	18.4237	22.9280	10.3757
srv_0	1	5	1415339	1.415339	47.5779	0.4046	18.7975	22.9112	10.3088
srv_0	1	6	1412452	1.412452	47.5057	0.4098	18.7701	22.9670	10.3474
compute_fp_1	1	1	1575316	1.575316	25.7445	28.8089	25.7444	19.7022	0.0000
compute_fp_1	1	2	1573994	1.573994	25.7059	28.8705	25.7056	19.7180	0.0000
compute_fp_1	1	3	1565764	1.565764	25.4635	29.2584	25.4635	19.8146	0.0000
compute_fp_1	1	4	841060	0.841060	19.5213	25.7144	3.2817	38.6254	12.8572
compute_fp_1	1	5	840492	0.840492	19.4822	25.7360	3.2683	38.6455	12.8680
compute_fp_1	1	6	840190	0.840190	19.4600	25.7477	3.2615	38.6569	12.8739
compute_int_0	1	1	1064985	1.064985	53.2393	0.8778	3.3183	27.0201	15.5445
compute_int_0	1	2	1089666	1.089666	52.7243	2.8805	4.4473	25.5281	14.4198
compute_int_0	1	3	1064988	1.064988	53.2393	0.8778	3.3184	27.0200	15.5445
compute_int_0	1	4	1064990	1.064990	53.2397	0.8778	3.3183	27.0196	15.5446
compute_int_0	1	5	1064987	1.064987	53.2393	0.8778	3.3184	27.0201	15.5444
compute_int_0	1	6	1064976	1.064976	53.2440	0.8721	3.3174	27.0199	15.5466
srv_0	2	1	1412629	1.177191	47.5764	0.4487	18.7246	22.9315	10.3188
srv_0	2	2	1385110	1.154258	47.9605	0.7024	18.2051	22.8653	10.2667
srv_0	2	3	1400337	1.166948	47.7866	0.5066	18.4880	22.8909	10.3279
srv_0	2	4	1393754	1.161462	47.6417	0.6309	18.4237	22.9280	10.3757
srv_0	2	5	1418785	1.182321	47.5779	0.4046	18.7975	22.9112	10.3088
srv_0	2	6	1415939	1.179949	47.5057	0.4098	18.7701	22.9670	10.3474
compute_fp_1	2	1	1863405	1.552838	25.7445	28.8089	25.7444	19.7022	0.0000
compute_fp_1	2	2	1862699	1.552249	25.7059	28.8705	25.7056	19.7180	0.0000
compute_fp_1	2	3	1858348	1.548623	25.4635	29.2584	25.4635	19.8146	0.0000
compute_fp_1	2	4	969810	0.808175	19.5213	25.7144	3.2817	38.6254	12.8572
compute_fp_1	2	5	969304	0.807753	19.4822	25.7360	3.2683	38.6455	12.8680
compute_fp_1	2	6	969029	0.807524	19.4600	25.7477	3.2615	38.6569	12.8739
compute_int_0	2	1	1073763	0.894802	53.2393	0.8778	3.3183	27.0201	15.5445
compute_int_0	2	2	1117626	0.931355	52.7243	2.8805	4.4473	25.5281	14.4198
compute_int_0	2	3	1073766	0.894805	53.2393	0.8778	3.3184	27.0200	15.5445
compute_int_0	2	4	1073768	0.894807	53.2397	0.8778	3.3183	27.0196	15.5446
compute_int_0	2	5	1073765	0.894804	53.2393	0.8778	3.3184	27.0201	15.5444
compute_int_0	2	6	1073697	0.894748	53.2440	0.8721	3.3174	27.0199	15.5466
srv_0	3	1	1835803	1.079884	47.5764	0.4487	18.7246	22.9315	10.3188
srv_0	3	2	1800349	1.059029	47.9605	0.7024	18.2051	22.8653	10.2667
srv_0	3	3	1820402	1.070825	47.7866	0.5066	18.4880	22.8909	10.3279
srv_0	3	4	1811804	1.065767	47.6417	0.6309	18.4237	22.9280	10.3757
srv_0	3	5	1842873	1.084043	47.5779	0.4046	18.7975	22.9112	10.3088
srv_0	3	6	1840527	1.082663	47.5057	0.4098	18.7701	22.9670	10.3474
compute_fp_1	3	1	1969360	1.158447	25.7445	28.8089	25.7444	19.7022	0.0000
compute_fp_1	3	2	1968354	1.157855	25.7059	28.8705	25.7056	19.7180	0.0000
compute_fp_1	3	3	1962056	1.154151	25.4635	29.2584	25.4635	19.8146	0.0000
compute_fp_1	3	4	1613460	0.949094	19.5213	25.7144	3.2817	38.6254	12.8572
compute_fp_1	3	5	1613319	0.949011	19.4822	25.7360	3.2683	38.6455	12.8680
compute_fp_1	3	6	1613257	0.948975	19.4600	25.7477	3.2615	38.6569	12.8739
compute_int_0	3	1	1572973	0.925278	53.2393	0.8778	3.3183	27.0201	15.5445
compute_int_0	3	2	1570238	0.923669	52.7243	2.8805	4.4473	25.5281	14.4198
compute_int_0	3	3	1572975	0.925279	53.2393	0.8778	3.3184	27.0200	15.5445
compute_int_0	3	4	1572968	0.925275	53.2397	0.8778	3.3183	27.0196	15.5446
compute_int_0	3	5	1572977	0.925281	53.2393	0.8778	3.3184	27.0201	15.5444
compute_int_0	3	6	1572965	0.925274	53.2440	0.8721	3.3174	27.0199	15.5466
srv_0	4	1	1838207	1.021226	47.5764	0.4487	18.7246	22.9315	10.3188
srv_0	4	2	1804301	1.002389	47.9605	0.7024	18.2051	22.8653	10.2667
srv_0	4	3	1823125	1.012847	47.7866	0.5066	18.4880	22.8909	10.3279
srv_0	4	4	1815511	1.008617	47.6417	0.6309	18.4237	22.9280	10.3757
srv_0	4	5	1845225	1.025125	47.5779	0.4046	18.7975	22.9112	10.3088
srv_0	4	6	1842908	1.023838	47.5057	0.4098	18.7701	22.9670	10.3474

compute_fp_1	4	1	2166383	1.203546	25.7445	28.8089	25.7444	19.7022	0.0000
compute_fp_1	4	2	2165529	1.203072	25.7059	28.8705	25.7056	19.7180	0.0000
compute_fp_1	4	3	2160202	1.200112	25.4635	29.2584	25.4635	19.8146	0.0000
compute_fp_1	4	4	1742032	0.967796	19.5213	25.7144	3.2817	38.6254	12.8572
compute_fp_1	4	5	1741999	0.967777	19.4822	25.7360	3.2683	38.6455	12.8680
compute_fp_1	4	6	1741996	0.967776	19.4600	25.7477	3.2615	38.6569	12.8739
compute_int_0	4	1	1580827	0.878237	53.2393	0.8778	3.3183	27.0201	15.5445
compute_int_0	4	2	1597350	0.887417	52.7243	2.8805	4.4473	25.5281	14.4198
compute_int_0	4	3	1580829	0.878238	53.2393	0.8778	3.3184	27.0200	15.5445
compute_int_0	4	4	1580822	0.878234	53.2397	0.8778	3.3183	27.0196	15.5446
compute_int_0	4	5	1580831	0.878239	53.2393	0.8778	3.3184	27.0201	15.5444
compute_int_0	4	6	1580768	0.878204	53.2440	0.8721	3.3174	27.0199	15.5466

Two Factor Analysis

	Depth			
Trace	1	2	3	4
srv_0	1.408889	1.177191	1.079884	1.021226
	1.379185	1.154258	1.059029	1.002389
	1.396131	1.166948	1.070825	1.012847
	1.388349	1.161462	1.065767	1.008617
	1.415339	1.182321	1.084043	1.025125
	1.412452	1.179949	1.082663	1.023838
compute_fp_1	1.575316	1.552838	1.158447	1.203546
	1.573994	1.552249	1.157855	1.203072
	1.565764	1.548623	1.154151	1.200112
	0.841060	0.808175	0.949094	0.967796
	0.840492	0.807753	0.949011	0.967777
	0.840190	0.807524	0.948975	0.967776
compute_int_0	1.064985	0.894802	0.925278	0.878237
	1.089666	0.931355	0.923669	0.887417
	1.064988	0.894805	0.925279	0.878238
	1.064990	0.894807	0.925275	0.878234
	1.064987	0.894804	0.925281	0.878239
	1.064976	0.894748	0.925274	0.878204

	Depth						
Trace	1	2	3	4	Row Sum	Row Mean	Row Effect
srv_0	1.400058	1.170355	1.073702	1.015674	4.659788	1.164947	0.085102
compute_fp_1	1.206136	1.179527	1.052922	1.085013	4.523598	1.130900	0.051054
compute_int_0	1.069099	0.900887	0.925009	0.879762	3.774756	0.943689	-0.136156
Column Sum	3.675292	3.250769	3.051633	2.980448			
Column Mean	1.225097	1.083590	1.017211	0.993483		1.079845	
Column Effect	0.145252	0.003744	-0.062634	-0.086362			

Overall Mean	1.079845
SSY	87.165160
SS0	83.95672853
SST	3.208432
SS_Trace	0.6812963384
SS_Depth	0.5848868815
SS_TraceDepth	0.026640
SSE	1.915609

Allocation of Variation

Pipeline depth (%)	Trace (%)	Interaction (%)	Error (%)
18.22968077	21.2345586	0.8303029256	59.7054577

Analysis of Variation

	F_Depth	F_Trace	F_Interact
MS/MSE	6.106537485	10.66965706	0.1390665037
Confidence level	0.95	0.95	0.95
f-value (table)	2.76	3.15	2.25
	significant	significant	no significant

18.23% of variation is explained by Depth config. Factor explains significant amout of variation at the 95% confidence level.									
21.23% of variation is explained by the Trace. Factor explains significant amout of variation at the 95% confidence level.									
0.83% of variation is explained by interaction. Factor doesn't explain a significant amount of variation at the 95% confidence level.									
59.71% of variation is explained by experimental error.									
Group Contribution									
Alicia Lam	33%	simulation code framework, debugging, testing & experimentation, analysis, report							
Tommy Truong	33%	code comments, skeleton & debugging, testing & experimentation, analysis, report							
Temirlan Utarbayev	33%	simulation implementation, testing & experimentation, analysis, report							